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Oefeningenreeks 2D nr. 16.3

$$\lim_{x \to \frac{\pi}{2}} \left( \frac{\cos x}{1 - \sin x} \right) = \lim_{x \to \frac{\pi}{2}} \left( \frac{\cos x(1 + \sin x)}{(1 - \sin x)(1 + \sin x)} \right) = \lim_{x \to \frac{\pi}{2}} \left( \frac{\cos x + \sin x \cos x}{1 - \sin^2 x} \right)$$

$$= \lim_{x \to \frac{\pi}{2}} \left( \frac{\cos x + \sin x \cos x}{\cos^2 x} \right) = \lim_{x \to \frac{\pi}{2}} \left( \frac{1 + \sin x}{\cos x} \right) = \frac{2}{0} = \infty$$

Bekijk linker en rechter limiet

$$\lim_{x \to \frac{\pi}{2}-} \left( \frac{1 + \sin x}{\cos x} \right) = +\infty$$

$$\lim_{x \to \frac{\pi}{2}+} \left( \frac{1 + \sin x}{\cos x} \right) = -\infty$$

# References